

Avish Jhalani

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EXPERIENCE

Global AI Technologies

Jan 2026 - June 2026

Role: Software Engineering Intern

Tech Stack: Python (NumPy, Pandas, Matplotlib), MySQL, Excel, Power BI

- Performed end-to-end analysis of a Q4 e-commerce dataset (**1,000,000+ transactions**) using Python (Pandas, NumPy, Matplotlib) and MySQL to identify root causes of a return rate spike from **12% to 19%**.
- Built an interactive Power BI dashboard from scratch — cleaned raw CSV data, modelled relationships, and designed visualizations — then presented findings directly to the reporting manager.
- Identified high-risk order segments: COD orders (**84.4% return rate**) and size-mismatch returns (**37% of total**); translated findings into three concrete operational recommendations projected to reduce returns by **7–11%**.

PROJECTS

Collab-Docs | React, NestJS, Yjs, Socket.io, PostgreSQL, Redis, Prisma, Docker

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- Engineered a real-time collaborative document workspace using **Yjs CRDTs** and **WebSockets**, deployed across Vercel, Render, and Supabase, achieving **327ms** average round-trip sync latency on cross-continent connections.
- Reduced PostgreSQL write overhead by **98%** by replacing per-keystroke writes with a custom **5-second debounced** state-save buffer.
- Cut network payload size by **90%+** by transmitting edits as compressed binary **Yjs deltas (30–60 bytes/frame)** instead of full document state, using a **Redis Pub/Sub** adapter to sync state across load-balanced instances.

BloodLink | Next.js, NestJS, PostgreSQL, Redis, Prisma, JWT, Nodemailer

[Repository](#) | [Link](#)

- Built a full-stack donor-matching platform (Next.js frontend, NestJS/Prisma backend) that geolocates and matches blood requests to eligible donors within a 10km radius using **PostGIS spatial queries** (ST_DWithin, ST_Distance).
- Accomplished verified reliability of the core donor-matching flow, as measured by an **11-case E2E suite** passing against a seeded PostgreSQL database, by writing Jest/Supertest integration tests covering auth, matching, confirmation, and fulfilment.
- Secured sessions using JWT stored in httpOnly, sameSite cookies, protecting against **XSS-based token theft**.

Notification Service API | NestJS, PostgreSQL, Redis, BullMQ, Prisma, Docker, JWT, Resend

[Repository](#)

- Built a production-ready REST API for async email delivery — API responds in **~300ms** by removing email sending from the request lifecycle using Redis job queues (**BullMQ**) and a background worker, with automatic retry via exponential backoff (**5s → 10s → 20s**) on failed deliveries.
- Secured all notification endpoints with JWT authentication (**7-day token expiry**) and enforced per-user rate limiting at **5 requests/minute**, blocking excess requests with 429 Too Many Requests.

TECHNICAL SKILLS

Programming Languages: C/C++, JavaScript, TypeScript, Python, SQL

Web & Cloud Technologies: Node.js, Express.js, React.js, NestJS, Next.js, RESTful API, Socket.io, AWS

Database Systems & Tools: PostgreSQL, Prisma, Redis, Git/GitHub, Postman, Docker

EDUCATION

JECRC University, Jaipur

2022-2026

Bachelor of Technology - Computer Science

CGPA: 8.7

- Coursework: Object-Oriented Programming, Data Structures, Algorithms, Operating Systems, Software Engineering, Database Management Systems, Computer Networks, System Design

ACHIEVEMENTS

- Solved over 500 problems across platforms, ranking in the top 14% in LeetCode Biweekly Contest 187
- Google Cloud Certified(Foundation).